

Professional Journey

A Narrative Through Academia, Industry, and Consulting

Mohsen Dirbaz
Chemical Engineer

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Professional Narrative — Mohsen Dirbaz

1 Foundation: Scientific & Academic Core

1.1 Formation: Petroleum and Natural Gas Engineering

My academic path began in petroleum engineering at the Sharif University of Technology in Iran, where I earned a full scholarship through a national examination that admits roughly the top one percent of approximately half a million applicants. Learning from the country's most distinguished professors provided rigorous training that shaped my technical foundation. As with many petroleum programs, the curriculum is inherently empirical relative to other disciplines; Sharif, in particular, sits at the high end of experimental emphasis. Consequently, I gained substantial hands-on experience in PVT laboratories, drilling laboratories, and reservoir rock analysis—training that grounded my understanding of hydrocarbon reservoirs and fluid mechanics.

The decision to pursue graduate study in the United States reflected the adventurous, opportunity-seeking spirit of my generation. While the practical range of choices proved narrower than initial hopes, I weighed two options in petroleum engineering—at the University of Alaska Fairbanks and at Texas A&M University–Kingsville (TAMUK). I chose TAMUK's pioneering Master's program in Natural Gas Engineering for its strong focus on process systems, which aligned with my evolving interests. To the best of my knowledge, TAMUK housed the first such program in the nation and now even offers a bachelor's degree in natural gas engineering.

During my master's program I became deeply engaged with process simulation and optimization. In advanced natural gas processing I learned HYSYS, which excels in phase behavior and gas-processing systems; in chemical processing and economics I used Aspen Plus, particularly well-suited to reactive systems and complex process design. In an advanced course on natural gas production and distribution, I completed projects such as compressor-station configuration for pipeline networks, selecting optimal pipe diameters, grades, and materials against pressure requirements and cost constraints.

By 2010 the shale boom had reshaped the energy landscape. I accordingly selected a thesis in that domain: *"Evaluation of Formation Damage in Horizontally Fractured Wells in Low-Permeability Reservoirs."* This work brought environmental dimensions of hydrocarbon production into sharp relief—insights that later influenced my doctoral research trajectory.

1.2 Advanced Research: Doctoral Studies in Chemical Engineering

Academia trains one to acknowledge limitations, question methods, and build incrementally on established knowledge. Success depends on rigorous peer review, reproducibility, and candid recognition of uncertainty. This ethos stands in marked contrast to commercial cultures that sometimes prize bold claims, rapid iteration, and market positioning over methodological soundness. Guided by the academic ethos and a desire to contribute substantively to scientific knowledge, I pursued doctoral studies in Chemical Engineering at the Illinois Institute of Technology.

The early phase of my program was demanding as I transitioned into advanced coursework. In Fall 2012 I received three B grades in foundational courses—Transport Phenomena, Applied Mathematics for Chemical Engineering, and Advanced Thermodynamics. Rather than treating these as setbacks, I used them as a catalyst. I fundamentally restructured my approach to graduate-level learning and adopted more effective study strategies. From Spring 2013 onward I earned consecutive A grades in all subsequent coursework, completing my PhD in 2020 with a GPA of 3.58.

The eight-year duration—longer than certain business-oriented programs that prioritize speed—afforded depth, breadth, and apprenticeship that shaped my research and teaching philosophies. I had the privilege of

weekly research meetings from the outset under mentors of notable integrity and acumen; I often remark, half in jest, that the “transfer of properties” applies to scholarly communities. The extended timeline also enabled extensive teaching assistantship experience—an opportunity many doctoral graduates do not enjoy—which strengthened my conviction in patient, methodical knowledge transfer and the importance of making complex concepts accessible.

During the exploratory phase I completed two stand-alone projects across distinct energy and environmental domains. First, over roughly six months, I studied the environmental impacts of hydraulic fracturing and assessed emerging options for produced-water remediation, including physico-chemical and biological treatment pathways and their process models. This project emerged organically while another student in my group examined fouling in water-treatment facilities; our efforts naturally converged. The work sharpened my sustainability perspective by foregrounding full life-cycle considerations rather than focusing solely on primary energy production.

Second, over approximately two years, I investigated gasification and emission-control technologies for Integrated Gasification Combined Cycle (IGCC) systems. I examined biomass and coal gasification across fixed, entrained, and fluidized-bed reactors, with emphasis on gasification and power-generation islands, and on intermediate purification steps—sulfur removal, trace-metal abatement, COS hydrolysis, and carbon capture—critical to technical and economic viability.

My dissertation developed a general predictive model for biomass gasification in fluidized-bed systems. The aim was to overcome limitations of purely kinetic or purely thermodynamic models by adopting strategies for complex process dynamics, sparse and heterogeneous experimental data, and incomplete solid-fuel characterization. I addressed data issues—missing values, irregular cardinality, and mass/energy-balance closure—and, using machine-learning predictive analytics, extracted features that earlier models had overlooked, particularly: structural components (cellulose, hemicellulose, lignin), proximate analysis parameters (volatile matter, fixed carbon, ash), and detailed ash composition.

Taken together, this trajectory shows how an extended doctoral period enabled exploration across independent domains—six months of environmental remediation research that shaped my sustainability outlook, two years on IGCC technologies, and finally a unifying step into advanced predictive modeling—forming a foundation that would have been difficult to build in an abbreviated program focused solely on time-to-degree.

1.3 Pedagogy & Leadership

Throughout graduate school I served as a teaching assistant for core chemical-engineering courses—Fluid Mechanics, Thermodynamics I and II, and Process Design I and II. I managed diverse student portfolios with speed, accuracy, and courtesy, maintaining an open-door policy to maximize availability and support.

Students elected me Teaching Assistant of the Year for 2015–2016 and again for 2016–2017 by votes of both undergraduate and graduate students. This recognition mattered to me because it suggested that I had cultivated effective communication despite initial concerns.

I also served as President of the Graduate Student Association in the Department of Chemical and Biological Engineering at IIT. The role entailed facilitating multi-stakeholder discussions, advocating for student needs with faculty and administration, and building collaborative solutions to departmental challenges—experience that developed my consensus-building and cross-functional communication.

As a member of the Students’ Ethics Committee, I helped organize workshops to establish discipline- and laboratory-specific ethical frameworks for research conduct. This initiative formed part of the NSF-funded project, “*A Bottom-Up Approach to Building a Culture of Responsible Research and Practice in STEM.*” My contributions earned commendation from the Director of the Center for the Study of Ethics in the Professions for effective leadership in advancing responsible research practices.

2 Applied Practice: Engineering & Strategy

2.1 Industrial Process Engineering (Larsen & Toubro)

I joined Larsen & Toubro Technology Services in 2019 through a timely networking opportunity during the final phase of my PhD. The role provided valuable industrial experience alongside the completion of my doctoral work.

As a process engineer, I reviewed process designs for chemical plants, verified heat and material balances, coordinated equipment specifications with vendors, and resolved discrepancies. Large projects required coordination across process, civil, mechanical, and electrical disciplines, each with its own constraints. Frequent cross-disciplinary meetings ensured that foundations supported equipment loads, piping and instrumentation were constructible, and electrical systems met specified power requirements. This work embedded the practical discipline of translating process design into constructible, operationally sound projects.

2.2 Strategic Advisory & Corporate Development (Didas International)

After completing my doctorate, I joined Didas International as a strategic advisor, building on a professional connection dating to my bachelor's internship. My recruitment coincided with the firm's strategic aim for expansion into renewable-energy markets and its need for sophisticated analysis to guide that transition.

Didas International—an established construction, mining, and infrastructure enterprise—maintains an extensive portfolio across Central Asia and North Africa, with numerous projects exceeding \$100 million in Iran, Kazakhstan, Armenia, Belarus, the UAE, and Algeria. I served as trusted counsel to senior leadership on strategic initiatives that would impact the company's position in emerging energy sectors.

My responsibilities extended beyond market scoping to the full arc of strategic planning: identifying viable business prospects, conducting detailed feasibility analyses of advanced technologies, and reporting to C-suite executives with actionable intelligence on high-stakes decisions requiring both technical depth and commercial judgment.

Engagements spanned interconnected domains. I led technical and economic evaluations of flare-treatment technologies, assessing regulatory frameworks and their implications for capital deployment. I also optimized silica-processing methodologies for solar-thermal applications—my first deep immersion into practical aspects of renewable-manufacturing—integrating materials-science requirements for high-performance reflective coatings with end-use performance criteria. Perhaps the most intellectually demanding project was a comprehensive economic-impact analysis of prospective Euro adoption scenarios in Middle-Eastern economies. That work combined international monetary policy, energy-trade dynamics, and macroeconomic modeling to assess how currency stability shapes long-term GDP patterns.

These experiences reinforced a strategic, collaborative approach to complexity—integrating rigorous technical analysis with commercial intelligence—and helped position the firm for transformation in rapidly evolving energy markets. The breadth of challenges provided perspective on the intersection of innovation, regulation, and business strategy—insight that later underpinned my work in techno-economic assessment and sustainable-energy systems.

3 Blossoming Innovation: Postdoctoral Fellowship

I accepted a postdoctoral position at the University of Missouri (2021–2025), which afforded me the opportunity to expand my horizons with research on emerging energy technologies in context of semiconductor industry.

3.1 TEA Space: Platform Architecture and Implementation

During this appointment, I developed *TEA Space*, an integrated software platform for techno-economic assessment that couples process modeling with rigorous economic analysis. The platform provides a standardized computational environment for baseline and investment-case comparisons across multiple project parameters, supporting both academic research and industrial decision-making workflows.

The implementation proceeded across three development phases:

Completed Components

- **Core cash-flow engine** performing base financial analysis across multiple scenarios, consolidating parameterized configurations into a unified financial horizon.
- **Visualization layer** supporting both dynamic (interactive) dashboards and static (publication-ready) plots for scenario comparison.
- **Temporal parameter control** with activation windows and level-based configurations enabling fine-grained scenario design.
- **Integrated cost console** for complex scaling and factorization of cost components across diverse parameter types.
- **Scenario management** including structured import/export for collaboration and version control.

In-Progress Development

- Migration to a **matrix-based computation layer** with advanced React state management for improved performance and UI responsiveness.

Planned Enhancements

- **Climate Intelligence Layer** integrating material-level carbon accounting and lifecycle footprint metrics, extending platform scope from techno-economic to techno-environmental-economic assessment.

3.2 Feasibility Study: Nitrogen Trifluoride (NF₃)

A principal assignment during the postdoctoral appointment was a comprehensive techno-economic feasibility study of nitrogen trifluoride (NF₃) for semiconductor applications. NF₃ is an ultra-high-purity electronic gas used primarily for chamber cleaning and selective etch. The study integrated: (i) resource geography and feedstock assurance, (ii) process-route definition and purification strategy, and (iii) capital and operating cost estimation with risk-sensitive scenarios.

Analytical Approach

The work proceeded in three phases designed to address major technical and commercial uncertainties:

1. **Upstream resources and materials assurance.** Survey and qualification of domestic fluorine pathways (fluorspar and fluorapatite value chains), beneficiation options, and exposure to import concentration and logistics variability.

2. **Synthesis and purification strategy.** Definition of process train alternatives, unit-operation integration, and contamination control to meet semiconductor-grade specifications; assessment of utilities intensity, hazardous-materials handling, and off-gas treatment requirements.
3. **Investment analysis.** Equipment list development to building-block level, capacity selection, CAPEX/OPEX stratification, and sensitivity analysis across power price regimes, feedstock quality variation, and by-product disposition.

3.3 Green Pathways: Hydrogen & Ammonia

To support the NF_3 study within realistic energy and feedstock contexts, I developed companion analyses of hydrogen (H_2) and ammonia (NH_3).

I analyzed steam-methane reforming, electrolysis, and biomass gasification, developing comparative frameworks across feedstock, efficiency, capital costs, and environmental impacts while accounting for technology maturity and regional resource availability.

Ammonia represents both a commodity chemical (primary use: fertilizers) and a potential energy vector. I examined production routes spanning conventional Haber–Bosch synthesis through emerging electrochemical and lithium-mediated pathways, requiring competency development in electrode materials, electrolyte systems, and reactor operation.

Ammonia (NH_3)

1. **Conventional (“Gray”) Haber–Bosch.** Baseline synthesis with process energy and carbon intensity benchmarks to establish reference costs and energy balances.
2. **“Blue” pathways with CO_2 capture.** SMR/ATR integration with CCUS, capture from concentrated and dilute streams, and impacts on levelized production costs, utilities, and plot-space.
3. **“Green” pathways.** Electrolytic hydrogen coupling and emergent electrochemical NH_3 routes; readiness levels, quantification protocols, and scale-up constraints.
4. **Transport, storage, and use.** Interface with hydrogen vectors and end uses (fertilizer, energy carrier), highlighting system couplings relevant to regional siting.

Hydrogen (H_2)

1. **Market and role.** Cross-sector integration and balancing services for variable renewables; implications for an NF_3 site’s utilities strategy.
2. **Production routes.** Fossil with/without CCUS, electrolysis technology families and deployment pipeline, and biomass-based options framed by resource logistics.
3. **Transport and storage.** Pipeline, shipping, cavern storage and purity/leakage considerations; alignment of transport–storage choices with plant load profiles.
4. **Comparative TEA.** Capital intensity and operating cost drivers versus carbon performance, providing boundary conditions for NF_3 power and reagent procurement scenarios.

4 Vision: Equitable Decarbonization Architecture

I developed a comprehensive policy paper examining how climate investment structures systematically produce inequitable outcomes while undermining decarbonization effectiveness. The work documents three converging failures: clean energy subsidies concentrating among highest-income quintiles despite nominal equity objectives, categorical technology framing that obscures lifecycle emission dependencies and enables incumbent rent-seeking, and erosion of foundational institutional capacity that prevents disadvantaged populations from participating in energy transitions regardless of program access.

The electric vehicle case study reveals the mechanism. "Zero emission" classification allows vehicles manufactured with fossil-intensive processes and charged on coal-dominated grids to receive identical subsidies and environmental credit as vehicles with genuinely low lifecycle emissions. This categorical framing benefits incumbent utilities, natural gas producers, and established manufacturers. Meanwhile, manufacturing emissions reducible by over ninety percent through available renewable-powered battery production, green steel, and responsible mining remain systematically unaddressed because all three actor classes—commercial, policy, civic—satisfy their objectives through symbolic classification rather than measured performance.

Biomass-derived hydrogen offers a contrasting pathway: lifecycle emissions depend directly on feedstock sustainability and process energy rather than external grid composition, preventing the opacity that enables three-actor evasion. My techno-economic analysis demonstrates that modular gasification clusters can achieve competitive levelized costs approaching fossil pathways with moderate carbon pricing while enabling distributed rural participation. Infrastructure requirements create barriers but also opportunities for distributed ownership models serving rural feedstock sources.

I propose policy architecture guardrails targeting benefit distribution rather than aggregate investment levels: mandatory lifecycle labeling eliminating categorical framing, subsidy scaling tied to demonstrated emission reductions rather than technology classification, infrastructure investment parity across pathways to enable comparative assessment, and deliberative allocation processes incorporating stakeholder representatives rather than technocratic optimization. Parliament-based structures resist incumbent capture; reusable capital structures enable multiple project cycles from single appropriations.

The analysis establishes fundamental physical constraints defining technology boundaries. Battery systems face electrochemical impossibility for applications requiring extended operational ranges, while hydrogen demonstrates exclusive viability for planetary-scale transportation through decades of operational precedent. The energy density advantage reflects categorical difference between electrochemical storage and chemical fuel systems—not incremental optimization opportunity but fundamental physics distinction determining viable pathways for freight-intensive applications serving mass populations.

My framework concludes that foundational institutional capacity—educational systems, digital infrastructure, economic opportunity—constitutes prerequisite rather than secondary consideration. Programs designed assuming functional educational systems, ubiquitous broadband, and economically stable populations will replicate existing inequities by creating opportunities accessible primarily to those already possessing advantages. Without structural countermeasures targeting benefit distribution, lifecycle transparency, and incumbent capture prevention, climate investment will reproduce rather than remediate systematic inequities documented across infrastructure domains.